

Practical no. 21

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Class :- CO3K – B

Subject :- DSU

Program :-

```
#include<stdio.h>
#include<conio.h>
#include<stdlib.h>
#define SIZE 5
int queue[SIZE];
int front = -1, rear = -1;
void insert(int value){
    if(rear==SIZE-1){
        printf("\nQueue Overflow! Cannot insert %d",value);
    }
    else{
        if(front==-1){
            front=0;
        }
        rear++;
        queue[rear]=value;
        printf("\n%d inserted into queue.",value);
    }
}
void delet(){
    if(front==-1 || front>rear){
        printf("\nQueue Underflow! Cannot delete.");
    }
    else{
        printf("\n%d deleted from queue.",queue[front]);
        front++;
    }
}
void display(){
    int i;
    if(front==-1 || front>rear){
        printf("\nQueue is empty.");
    }
    else{
        printf("\nQueue elements are :- ");
        for(i=front;i<=rear;i++){
            printf("\n%d",queue[i]);
        }
    }
}
```

```

printf("\n");
}
}
void main(){
int choice,value;
clrscr();
while(1){
printf("\n----LINEAR QUEUE MENU----");
printf("\n1.Insert\n2.Delete\n3.Display\n4.Exit");
printf("\nEnter your choice :- ");
scanf("%d",&choice);
switch(choice){
case 1:
printf("\nEnter value to insert :- ");
scanf("%d",&value);
insert(value);
break;
case 2:
delet();
break;
case 3:
display();
break;
case 4:
exit(0);
default:
printf("\nInvalid Choice!");
}
}
getch();
}

```

OUTPUT :-

```
DOSBox 0.74, Cpu speed: max 100% cycles

----LINEAR QUEUE MENU----
1.Insert
2.Delete
3.Display
4.Exit
Enter your choice :- 1

Enter value to insert :- 10

10 inserted into queue.

----LINEAR QUEUE MENU----
1.Insert
2.Delete
3.Display
4.Exit
Enter your choice :- 1

Enter value to insert :- 20

20 inserted into queue.

----LINEAR QUEUE MENU----
1.Insert
2.Delete
3.Display
4.Exit
Enter your choice :- 1

Enter value to insert :- 30

30 inserted into queue.

----LINEAR QUEUE MENU----
1.Insert
2.Delete
3.Display
4.Exit
Enter your choice :- 3

Queue elements are :-
10
20
30

----LINEAR QUEUE MENU----
1.Insert
2.Delete
3.Display
4.Exit
Enter your choice :- 2

10 deleted from queue.
```